Monthly Intelligence Report

Edition 001 — The Atacama–Central Transmission Corridor Corridor

Where demand surges are outrunning transmission capacity. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

1,095

Major + distribution across SEN (Sistema Eléctrico Nacional)

GRID LINES MAPPED

~2,081

~38,784 km transmission & distribution

MC SIMULATIONS

85.0 M

10,000 per substation

PUBLIC DATA SOURCES

20+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes 146,000+ source data points per run, executes 85.0 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 1,095 substations and ~2,081 grid lines spanning ~38,784 km.

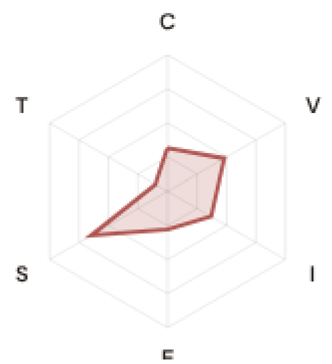
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 20+ verified public sources (CEN, CNE, INE, DMC, SERNAGEOMIN, CSN, ACERA), making the methodology fully auditable and reproducible. Initially covering Chile's SEN (SIC and SING) and isolated systems, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Subestación Cholguán (STS)

Biobío · Biobío · 220 kV

0.555	High	0.275	89th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- **C** Continuity: 0.32 / 0.30 (32%)
- **I** Infrastructure: 0.37 / 0.25 (37%)
- **S** Saturation: 0.65 / 0.20 (65%)
- **V** Voltage: 0.48 / 0.10 (48%)
- **E** Economic: 0.28 / 0.10 (28%)
- **T** Transition: 0.10 / 0.05 (10%)

MODIFIERS

- **R3 Consequence:** 1.027 ↑ amplifies
- **R4 Graph Criticality:** 1.008 → neutral
- **R6a Restoration:** 1.023 ↑ amplifies
- **R6b Seismic/Hazard:** 1.352 ↑ amplifies
- **R7 Digital Readiness:** 0.974 ↓ dampens

This month's spotlight — automatically selected for April 2026 — is **Subestación Cholguán (STS)** in Biobío, Biobío. With an R_final of 0.555 (High band, 89th fleet percentile), its risk profile is dominated by the V (Voltage) component at 481% of its weight ceiling, followed by S (Saturation) at 325%. Modifiers collectively shift R_base by 39.5, reflecting the substation's specific geographic, socio-economic, and network context. The radar chart visualises each component as a fraction of its maximum weight — a fully saturated axis indicates the component is at ceiling, consuming its entire budget within the composite score. This substation's profile illustrates the SSI's core principle: risk is never mono-dimensional. Even when one component dominates, the interaction of all six — modulated by the five multipliers — determines the final score that drives investment prioritisation.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Chile's Energy Security Board and network operator strategic planning requires.

CYBER HIGH EXPOSURE	BLIND SPOTS	AVG R7 MODIFIER
0	49	0.9802
0.0% of fleet	High/Critical + R7 < median	Range [0.99, 1.05]

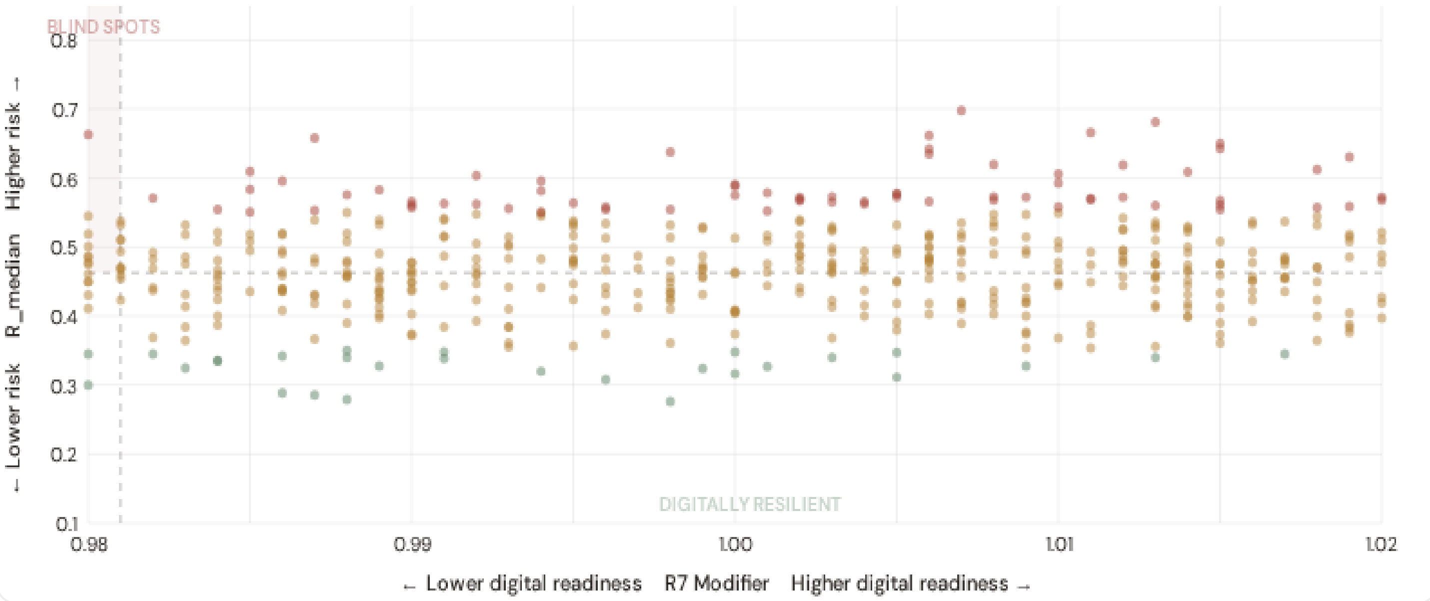
HIGH-CONSEQUENCE SUBS

0

0.0% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **49 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 0.9810). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Antofagasta (20), Biobío (15), O'Higgins (4)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating robust digital infrastructure coverage concentrated in metropolitan networks (Enel Distribución, CGE, Chilquinta).

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: CL$20–40/kWh

Copper/lithium mining, smelting, major industrial zones (Antofagasta, Atacama) — highest economic loss per interruption

275 substations (25.1%) High/Critical: **67** (24.4%) Avg R: **0.494** Avg E: **0.371** / 0.10

Low 13 Med 195 High 67 Crit 0

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: CL$10–20/kWh

Copper processing, forestry/cellulose, fishing industry, wine production — significant continuity requirements for export-driven operations

273 substations (24.9%) High/Critical: **32** (11.7%) Avg R: **0.470** Avg E: **0.374** / 0.10

Low 8 Med 233 High 32 Crit 0

🏢 Commercial / SME-Dense

Est. VoLL: CL$4–10/kWh

Santiago commercial district, Valparaiso port, regional service centres — moderate individual impact but high aggregate exposure across urban centres

281 substations (25.7%) High/Critical: **25** (8.9%) Avg R: **0.454** Avg E: **0.377** / 0.10

Low 23 Med 233 High 25 Crit 0

🌾 Light / Agricultural / Remote

Est. VoLL: CL$1–4/kWh

Smallholder agriculture, remote comunas, Patagonian settlements — lower VoLL but higher social vulnerability, longer restoration times, and isolation risk

266 substations (24.3%) High/Critical: **15** (5.6%) Avg R: **0.432** Avg E: **0.364** / 0.10

Low 43 Med 208 High 15 Crit 0

Value of Lost Load (VoLL) context: CEN's 2024 Value of Customer Reliability (VCR) report places average VoLL at CL$33.46/kWh for commercial/industrial and CL$19.37/kWh for residential customers across the SEN. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **58 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Antofagasta (49), O'Higgins (33), Metropolitana (30)** — regions where industrial corridors depend on uninterrupted power for continuous processes (aluminium smelting, LNG liquefaction, mining beneficiation). A single hour of unplanned outage at these substations carries an estimated VoLL of CL$20–40/kWh, compared to CL$1–4/kWh in remote pastoral areas. The fleet-wide average energy poverty rate is 12.3%, but this masks sharp state-level variation: South Chile and Los Ríos combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 State & Comuna Digital Readiness Ranking

comunas ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_median to identify regions combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 COMUNA REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Los Ríos	0.9695
2	Magallanes	0.9699
3	Valparaíso ⚠ high R	0.9734
4	Tarapacá ⚠ high R	0.9753
5	Araucanía	0.9784
6	Aysén	0.9787
7	Arica y Parinacota ⚠ high R	0.9800
8	Metropolitana	0.9800
9	Biobío ⚠ high R	0.9804
10	Antofagasta ⚠ high R	0.9805
11	Coquimbo	0.9817
12	Los Lagos	0.9817
13	Ñuble ⚠ high R	0.9818
14	Maule ⚠ high R	0.9823
15	Atacama	0.9825

comuna-level analysis reveals a clear digital divide mirroring Chile's metropolitan-regional connectivity gap. **Los Ríos** has the lowest average R7 (0.9695), while **O'Higgins** leads at 0.9844 — a spread of 0.0149 across the R7 range. **5 comunas** combine below-median digital readiness with above-median grid risk, making them priority targets for SEC/distribution company digitalisation investment and the Chilean Government's green energy programme. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open broadband/smart meter data — as ACSC/regulatory compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (April 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9802, median = 0.9810; 0 substations classified HIGH cyber-exposure; 49 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging comunas where broadband and smart meter rollouts (SEC/distribution company Advanced Metering Infrastructure programmes) translate into measurable R7 shifts. The next material R7 update is expected when INE publishes updated Digital Inclusion Index data (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 162 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

30

162 variables

WEEKLY / MONTHLY

9

High-frequency feeds

QUARTERLY / ANNUAL

17

Regular refresh cycle

STATIC / DERIVED

4

Standards + scenarios

Data Source Registry — April 2026

Loading source registry...

C.1 Update Frequency Timeline

Loading...

C.2 Fleet Score Snapshot

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C.3 Changelog — v3.4 → v4.0.2

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SECTION D — MONTHLY DEEP-DIVE

The Atacama–Central Transmission Corridor Corridor: Where Demand Surges Outrun Transmission Capacity

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Atacama–Central Transmission Corridor corridor · **002** South Chilean DER Saturation · **003** Atacama Extreme Heat Infrastructure Stress · **004** Metropolitan-Remote economic impact disparity · **005** Energy transition risk (Mining regions) · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Comuna Grid Health Cards · **010** Academic validation and peer feedback · **011** Latin American replication feasibility · **012** Year in review.

D.1 Why the Atacama–Santiago Corridor, Why Now

ATACAMA–CENTRAL SUBSTATIONS

195

111 HV · 84 MV

HIGH BAND

51

26.2% of corridor

CORRIDOR MEDIAN R

0.495

Fleet median: 0.463

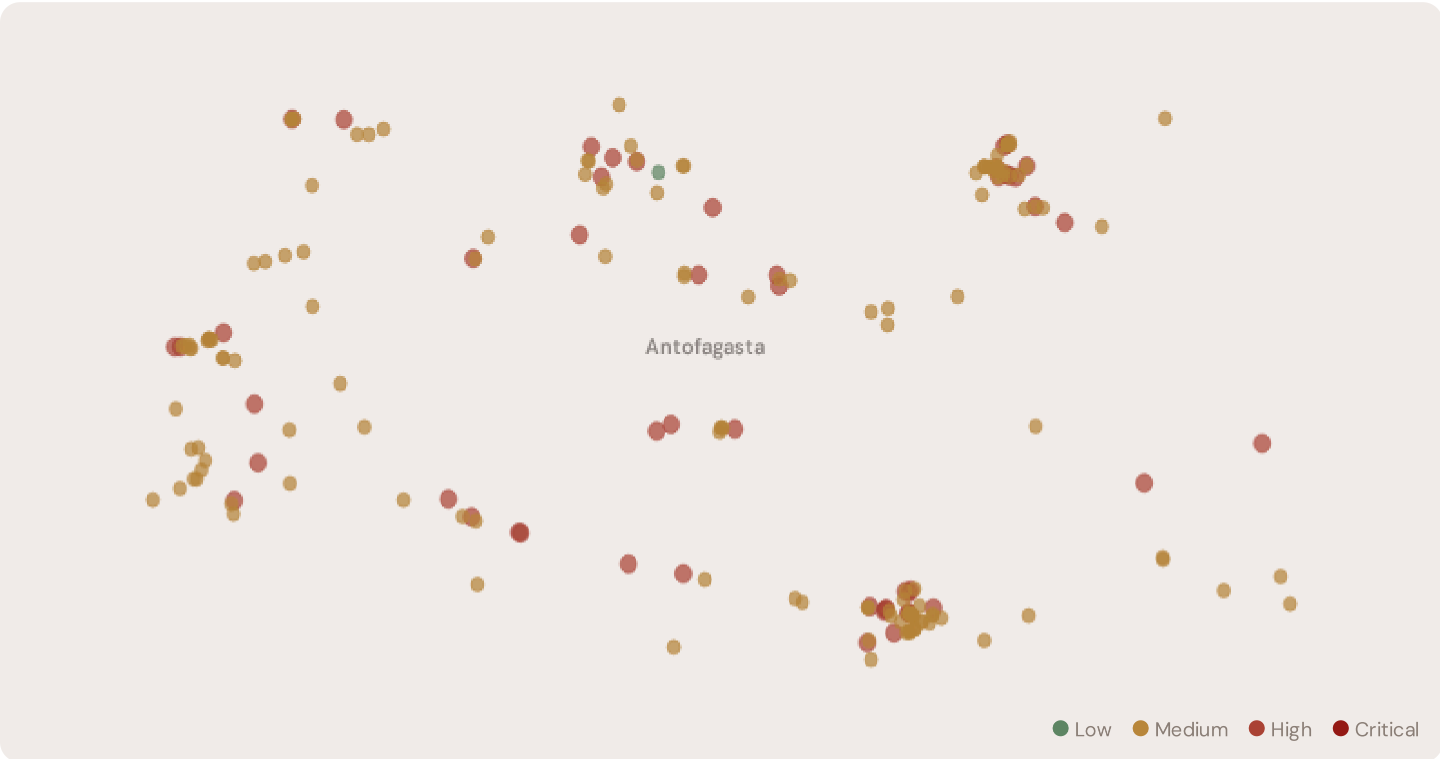
WORST COMUNA REGION

Antofagasta

Avg R: 0.500 · 195 substations

The Atacama–Central corridor hosts **195 substations** across multiple comunas, forming the backbone of Chile's east-coast interconnected grid. Its risk profile is disproportionately concentrated in the upper bands: **51 substations (26.2%)** are classified High, with only **1** in the Low band. 69% of Antofagasta substations score above the national fleet median of 0.463. The corridor median R of 0.495 reflects the tension between rapid coal-to-renewable transition, accelerating DER penetration, growing interconnector demand, and ageing infrastructure built for centralised baseload generation. The SSI flags continuity-of-supply deficits, infrastructure age mismatches, and the strain of bidirectional flows on a unidirectional network.

D.2 Corridor Map and Scoring



SUBSTATION	COMUNA REGION	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Subestación ANT 922	Antofagasta	0.698	High	0.603	0.223	0.518	0.439	0.496	0.719	T

SUBSTATION	COMUNA REGION	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Subestación ANT 894	Antofagasta	0.672	High	0.587	0.382	0.488	0.533	0.485	0.661	T
S/E Mejillones	Antofagasta	0.658	High	0.395	0.437	0.444	0.401	0.494	0.743	T
Subestación Uribe	Antofagasta	0.650	High	0.503	0.370	0.392	0.559	0.408	0.508	
Tap Off Llanos	Antofagasta	0.638	High	0.569	0.390	0.438	0.513	0.496	0.485	
Subestación Mina Ministro Hales	Antofagasta	0.638	High	0.560	0.414	0.500	0.525	0.423	0.414	
Subestación 415	Antofagasta	0.626	High	0.602	0.227	0.478	0.508	0.501	0.463	
Tap Off Oeste	Antofagasta	0.620	High	0.607	0.438	0.408	0.486	0.377	0.435	
Subestación ANT 840	Antofagasta	0.620	High	0.560	0.402	0.520	0.511	0.477	0.445	
Tap Off 003	Antofagasta	0.620	High	0.497	0.410	0.468	0.473	0.442	0.622	

… 185 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — ATACAMA–CENTRAL CORRIDOR VS FLEET

C — Continuity 0.447 vs 0.448 (-0%)

I — Infrastructure 0.405 vs 0.329 (+23%)

S — Saturation 0.417 vs 0.474 (-12%)

V — Voltage 0.348 vs 0.350 (-1%)

E — Economic 0.496 vs 0.372 (+34%)

T — Transition 0.571 vs 0.197 (+190%)

MODIFIER AVERAGES — ATACAMA–CENTRAL CORRIDOR VS FLEET

R3 Consequence	1.001	fleet 0.998	→
R4 Graph Crit.	0.980	fleet 0.980	↓
R6a Restoration	1.018	fleet 1.021	↑
R6b Climate Hazard	NaN	fleet NaN	→
R7 Digital Read.	0.980	fleet 0.980	↓

The dominant risk driver across the Atacama–Central corridor is **T (Transition)**, averaging 0.571 against a fleet average of 0.197 — occupying 1142% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.496 vs fleet 0.372 (496% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.001 (fleet: 0.998), reflecting the economic significance of Atacama–Central's industrial base — from copper to lithium processing mining logistics corridors. The R6b climate hazard modifier averages NaN, capturing the corridor's increasing exposure to extreme heat events and seismic hazard risk — a dimension invisible to every competing framework.

D.4 Comuna Region Breakdown

COMUNA REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Antofagasta 195 subs · avg R = 0.500 · H:51 M:143

The comuna-level breakdown reveals significant intra-regional variation. **Antofagasta** leads with a mean R of 0.500 across 195 substations, while **Antofagasta** scores 0.500 — a spread of 0.000 within a single state. This disparity underscores why state-level metrics (as used by CNE performance reports) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that Atacama–Central is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that CEN's Integrated System Plan and CEN capital expenditure planning requires.

D.5 Implications

4 Antofagasta substations score R ≥ 0.650, placing them in the upper quartile of national risk. For CEN ISP investment prioritisation and green energy funding allocation, this analysis identifies the Antofagasta corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by copper export dependency, growing copper mining demand, and seismically-active transmission corridors — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

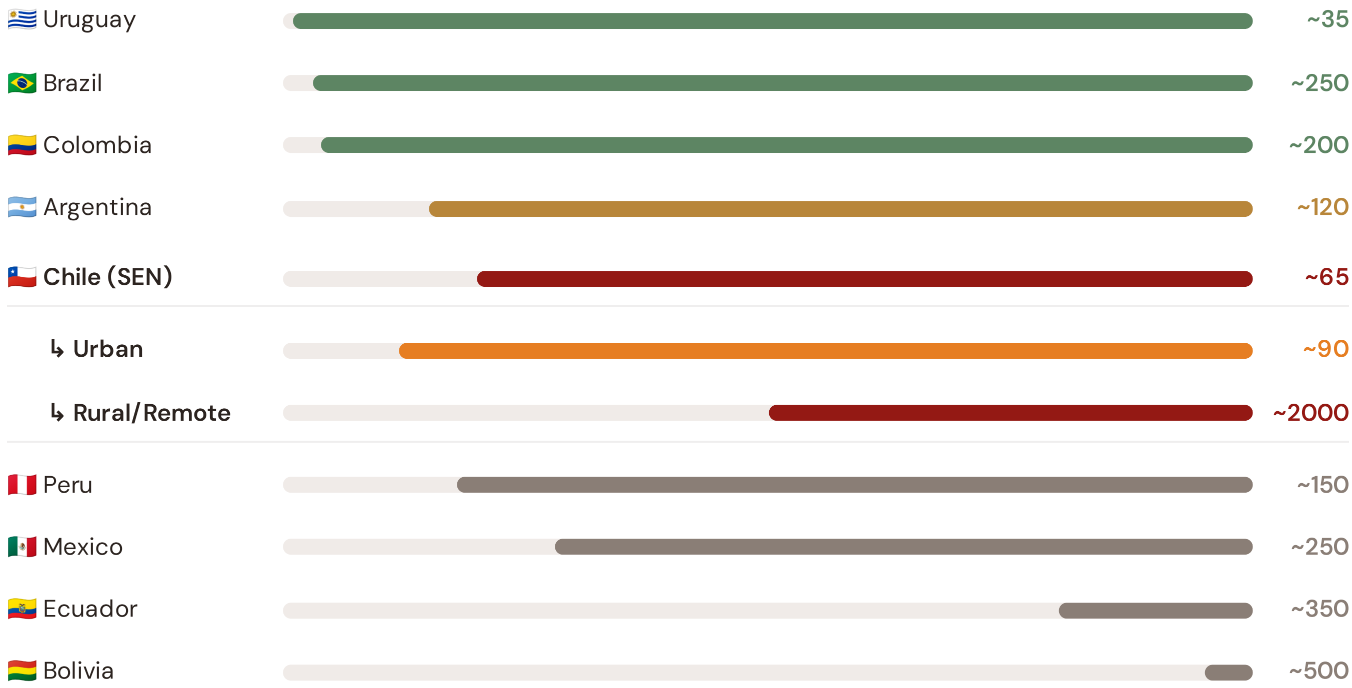
SECTION E — RECURRING MODULE

Latin American Context Card

How this works: Each edition includes one Latin American Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Atacama–Santiago corridor analysis hinges on Chile's continuity performance relative to regional peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Japan (003), CMIP6 climate hazard vs SE Asian peers (004), etc. The underlying data updates annually with CEN/CNE and national regulator publications.

Unplanned SAIDI (min/customer/year) — Selected Latin American Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



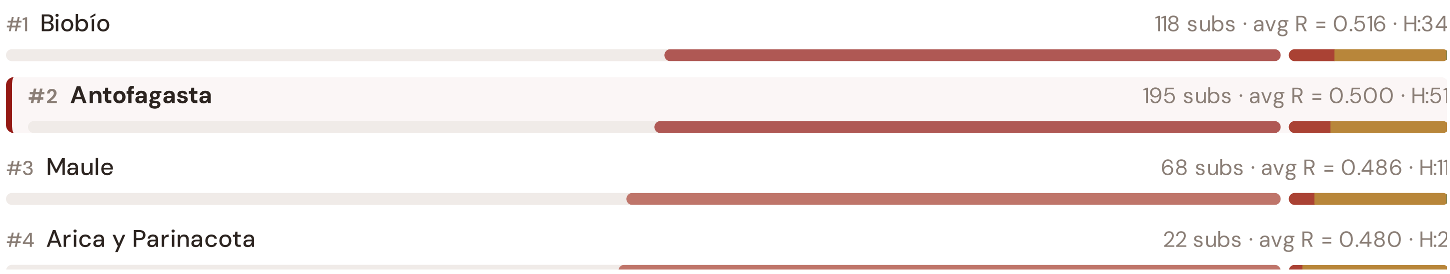
Source: CEN Reliability Panel Annual Report (2024); CNE State of the Energy Market (2025); World Bank ESMAP (2023); individual national regulator publications. Chile sub-categories from CNE urban/rural/remote classifications. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually.

This month's key insight: The Atacama–Central corridor deep-dive shows **195 substations** (of 195) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Chile's rural/remote territory classification. The Atacama–Central corridor's average C of 0.447 is **1.0× the fleet average** of 0.448. In Latin American terms, these substations individually perform closer to Malaysian or Thai SAIDI levels (~150–115 min/customer/year) than to Chile's own SEN urban average (~90 min). The SSI reveals what aggregate CNE statistics conceal: that Chile's mid-tier Latin American position is not a national condition but a statistical average of Singapore-quality urban performance and developing-nation-level remote performance — and the Atacama–Central corridor concentrates significant pockets of the latter.

State/Territory Risk Ranking — All 16 Chilean Regiones & Territories

Where does the Atacama–Central corridor sit within the national landscape? States sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



#5 Tarapacá — 47 subs · avg R = 0.479 · H:6

#6 Valparaíso — 59 subs · avg R = 0.479 · H:7

#7 Ñuble — 40 subs · avg R = 0.468 · H:1

#8 O'Higgins — 120 subs · avg R = 0.467 · H:12

#9 Atacama — 68 subs · avg R = 0.452 · H:5

#10 Coquimbo — 77 subs · avg R = 0.438 · H:3

#11 Araucanía — 22 subs · avg R = 0.434 · H:1

#12 Metropolitana — 142 subs · avg R = 0.433 · H:6

#13 Los Ríos — 28 subs · avg R = 0.413 · H:0

#14 Los Lagos — 62 subs · avg R = 0.381 · H:0

#15 Aysén — 11 subs · avg R = 0.320 · H:0

#16 Magallanes — 16 subs · avg R = 0.308 · H:0

Antofagasta ranks **#2 of 16 regiones** by mean R_median. The three most stressed states are Biobío, Antofagasta, Maule, while the three most resilient are Magallanes, Aysén, Los Lagos. 8 of 16 states score above the fleet average of 0.463. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate SEC/distribution company or national statistics — reveals the true spatial distribution of grid stress across Chile. It is precisely this substation-level granularity that positions the SSI as a tool for CEN ISP / green energy / CER investment prioritisation: not treating entire states as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the SEC/CEN and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	SEC/CEN + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering an CL$15M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down.

KEY CONCEPTS

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall visualisation: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The Chilen training set comprises 1,095 substations with complete 95-feature vectors. Average CI_width of 0.136 provides the uncertainty ground truth for neural network calibration. The fleet's risk distribution — 87 Low, 869 Medium, 139 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

Extreme South Remote Communities

Deep-dive into Punta Arenas's grid risk profile — substation map, component analysis, Comuna breakdown. Latin American Context Card: Remote community power supply and renewable integration.

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Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an CEN participant, distribution company, transmission operator, regional government, research institution, or industry body and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 1,095 substations · ~2,081 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5
